

**BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI**  
(MID-SEMESTER EXAMINATION MO/2023)

CLASS: BTECH  
BRANCH: CSE/AI ML/ECE/EEE

SEMESTER : I  
SESSION : MO/2023

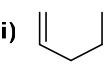
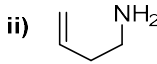
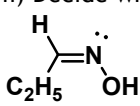
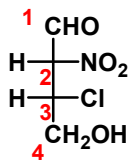
SUBJECT: CH101 CHEMISTRY

TIME: 02 Hours

FULL MARKS: 25

**INSTRUCTIONS:**

1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | CO | BL |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----|----|
| Q.1(a) What do you mean by primary and secondary valencies of metal in a coordination complex? [2]                                                                                                                                                                                                                                                                                                                                                                                                                |  | I  | 1  |
| Q.1(b) A coordination compound with formula $\text{CoCl}_3(\text{NH}_3)_4$ does not liberate $\text{NH}_3$ but precipitates one $\text{Cl}^-$ ion as $\text{AgCl}$ . Write the structural formula of compound and predict the molar conductivity. [3]                                                                                                                                                                                                                                                             |  | I  | 3  |
| Q.2(a) Calculate crystal field stabilization energy (CFSE) for the complex, $[\text{Co}(\text{CN})_6]^{3-}$ [2]                                                                                                                                                                                                                                                                                                                                                                                                   |  | I  | 2  |
| Q.2(b) $[\text{Co}(\text{NH}_3)_6]^{3+}$ is diamagnetic and orange-yellow. $[\text{CoF}_6]^{3-}$ on the other hand is paramagnetic and blue. Why? [3]                                                                                                                                                                                                                                                                                                                                                             |  | I  | 2  |
| Q.3(a) What are the criteria to have aromaticity in the compound? Discuss with an example of benzene. [2]                                                                                                                                                                                                                                                                                                                                                                                                         |  | II | 2  |
| Q.3(b) What is the number of $\pi$ electrons in the following molecules? With explanation. [3]                                                                                                                                                                                                                                                                                                                                                                                                                    |  | II | 2  |
| <div style="display: flex; justify-content: space-around; align-items: flex-end;"> <div style="text-align: center;"> <p>i) </p> </div> <div style="text-align: center;"> <p>ii) </p> </div> <div style="text-align: center;"> <p>iii) </p> </div> </div> |  |    |    |
| Q.4(a) i) What type of structural isomerism ethyl acetoacetate ( $\text{CH}_3\text{COCH}_2\text{COOC}_2\text{H}_5$ ) Show? [1+1]                                                                                                                                                                                                                                                                                                                                                                                  |  | II | 2  |
| ii) Decide whether the structure drawn is the E or Z isomer.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |    |    |
|                                                                                                                                                                                                                                                                                                                                                                                                                                |  |    |    |
| Q.4(b) (i) Identify the correct R/S configuration of C2 and C3 from the drawn structure. ii) Draw the correct sawhorse projection of the following structure. [2+1]                                                                                                                                                                                                                                                                                                                                               |  | II | 3  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                |  |    |    |
| Q.5(a) Why, in general, boiling points of cis-isomers are higher compared to trans-isomers? [2]                                                                                                                                                                                                                                                                                                                                                                                                                   |  | II | 2  |
| Q.5(b) Which one of the $\text{Ni}^{2+}$ or $\text{Cu}^{2+}$ complexes shows Jahn-Teller distortion & why? [3]                                                                                                                                                                                                                                                                                                                                                                                                    |  | I  | 3  |

:17/10/2023: